BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 040 · Programming Techniques · Functions and Procedures

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is the final result of x=5; x-=2;?

Q2. What does a binary search need first?

Q3. How many times runs for(i=0;i<4;i++)?

Q4. What does a truth table list?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

Recall the term — function

Recall the term — procedure

Recall the term — parameter

Recall the term — argument

Recall the term — return

New learning

Explain the important decision — static int Double(int value) { return value * 2; } Console.WriteLine(Double(6)); outputs 12.

Worked example

Return a computed value

1. Identify the task — Return a computed value

2. Follow the logic — static int Double(int value) { return value * 2; } Console.WriteLine(Double(6)); outputs 12.

3. Check the outcome — Explain how each step contributes to the result. Test an edge case.

We Do · Practise with guidance

Problem 1

A function adds a and b; what is Add(3, 4)?

Discuss, then respond

Check your thinking

7.

Problem 2

A void method prints a greeting. Does it return a string?

No: void has no return value.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Apply independently
Write a C# method returning the cube of an integer. [4]

Question 2

Apply independently
Explain the purpose of a parameter. [2]

Question 3

Apply independently
What is the value of Twice(Twice(2))? [2]

Exit Ticket · Answer before checking

True or false: a void C# method must return a calculated value.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. 3

Q2. Sorted input

Q3. 4

Q4. Outputs for input combinations

You Do pages · Independent questions

Q1. static int Cube(int n) { return n*n*n; }

Q2. It receives input for use inside a subroutine.

Q3. 8.

Exit Ticket · Model answer

False. A void C# method does not return a value.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: